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Project Report
On

“RAILWAY TRACK CRACK & OBJECT DETECTION USING IOT”

Submitted in the partial fulfillment for the award degree of

BACHELORS IN ENGINEERING

in

ELECTRONICS & COMMUNICATION ENGINEERING

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING CERTIFICATE

It is certified that the Project work entitled “**RAILWAY TRACK CRACK & OBJECT DETECTION USING IOT**” carried out by **Mr. RAVI KUMAR CHANDRA (1BI20EC111)** and **Ms. SUPRIYA RANI (1BI20EC156)** bonafide student of **Bangalore Institute of Technology** in partial fulfillment for the award of **Bachelor of Engineering in Electronics and Communication Engineering** of the **Visvesvaraya Technological University**, Belagavi during the year 2023-2024. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the Report deposited in the departmental library. The Project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the said degree.

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BANGALORE INSTITUTE OF TECHNOLOGY

VISION

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- Develop the Institute as a leader in Science, Engineering, Technology and management, Research and apply knowledge for the benefit of society.
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PROGRAM SPECIFIC OUTCOMES

PSO1:

Core Engineering: The graduates will be able to apply the principles of Electronics and Communication in core areas.

PSO2:

Soft Skills: An ability to use latest hardware and software tools in Electronics and Communication engineering.

PSO3:

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ABSTRACT

Most of the train accidents occur due to the breakage in track and the animal passage in the railway track (example : elephant , cow ,etc...). There are many methods used by Indian railway to avoid train accidents. But Indian railway struggles a lot to avoid it. Techniques like Automatic train protection system, Ultrasonic flaw detection system etc. In order to overcome these problem, we are using the few techniques to detect the breakage in track and detecting animals or any obstacle like small stones and rebars before entering the track. The sensors like Ultrasonic, Infrared sensor are used in the detection and prevention process. The IR sensor is used in the detection of crack in tracks. The ultrasonic is used in the detection of animals and obstacles .It is controlled by the Arduino mega and application software.

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CHAPTER 1

INTRODUCTION

1.1 Overview

Depending on the fast developments in railway systems, high-speed trains are used for speed transportation, and rail transportation is increased day by day. The most of the people uses railway for transportation because it is essential for transferring the goods and passengers from one place to another place and low cost. And also, the railway system is providing facility such as high speed, with economical, environment friendly, safety, and better characteristics of railway systems. These can be done by time to time maintenance and control measurements. But depending on different factors, deformations and derailment may occur on the superstructure of railways, because of improper maintenance and the currently irregular and manual track line monitoring mistake from workers. Such deformation is determining on time and taking precautions is very important for the safety of railway systems.

The unauthorized infiltration of animals, vehicles, and sometimes, humans affect the overall situation immensely by not only hampering the regular operation of trains but also being the reason for huge damage to their own lives as well as the surroundings. Considering all these scenarios, the most suitable solution that might reduce the overall accountability of humans in this regard can be video monitoring which is proven to be influential thanks to its wide range of applications. However, the traditional video monitoring system needs to use human labour as it tends to monitor the concurrent situation of the tracks manually. As per the necessity, it is required to construct a railway obstacle monitoring system based on machine learning algorithms, in which we can trust the machine to make decisions based on its own intelligence, allowing us to avoid the drawbacks of relying on human observations.

Therefore, solution for this problem is introducing in this project. To provide the protection to the railway accident because of cracks occurs in the railway track. This system is used in-between two stations which will detect the cracks present on the track using IR sensors which transmit sine waves for an ideal track. If a crack is detected then this sensor will send a signal to the Arduino Uno board which will activate the GPS receiver. The GPS receiver will send the exact location which will then be messaged to the main authorities. Using this smart technology will be a part of the brave new digitalized world which will be able to prevent the loss of precious life and property.

1.2 Motivation

The motivation to develop a project on railway track crack detection and object detection stems from the critical need to enhance railway safety and efficiency. Railway systems are vital for transportation globally, but inadequate maintenance and security threats pose significant risks, leading to derailments and accidents. By implementing automated detection systems, we can proactively identify track defects like cracks, minimizing the likelihood of derailments and ensuring passenger safety. Additionally, integrating object detection technology can help address security concerns by identifying and mitigating risks posed by unauthorized individuals or objects on railway premises. Ultimately, these projects aim to save lives, prevent accidents, and improve the overall reliability of railway systems.

1.3 Problem Statement

The primary issue in rail track maintenance is the lack of affordable and efficient tools for detecting problems, leading to cracks and derailments. This is compounded by inadequate maintenance practices and threats from anti-social activities.

Our model aims to address these challenges by implementing cost-effective, automated track assessment technologies. These technologies will enable real-time detection and prompt addressing of track issues, such as cracks, which are major causes of derailments. The implementation will involve advanced detection systems, improved maintenance protocols, and enhanced security measures to mitigate threats, thereby improving rail safety and reliability, particularly in high-incidence areas like India.

CHAPTER 2

LITERATURE SURVEY

2.1 Technical Papers

- [1] **“Efficient Monitoring System for Railways for Crack Detection (IEEE), Manisha Vohra, S.K. Gabhane, 2022”**

Railways serve people with numerous benefits like helping in reaching places very quickly, providing low priced fare traveling. The railways provide one of the most largely used modes of public transportation. Nowadays, in Indian railways, a frequent number of accidents have taken place due to faults like the presence of cracks on rail tracks. To overcome this problem, we are using IoT-based railway track crack detection.

- [2] **“Design and Implementation of Wireless Security System for Railway Tracks, Mehaboob. Mujawar, Sangam.Borkar, 2021”**

This paper describes an electronic system that can help detect the exact location of the crack. In India, railway transportation plays a crucial role in connecting the entire country via different routes in hilly regions, deserts, plateaus, and other extreme climatic conditions. We are implementing a system that can locate the exact location of the crack on the track, helping to reduce rail accidents, which often lead to heavy loss of life and property.

- [3] **“Autonomous Railway Crack Detector Robot for Bangladesh: SCANOBOT, Nagib Mahfuz, Omor Ahmed Dhali, Safayet Ahmed, Mehen Nigar, 2021”**

This paper explains an automatic railway track crack detection system using a robot. This robot includes two ultrasonic sensors, GPS, GSM modules, and Arduino Mega-based crack detection, which is cost-effective and facilitates better safety standards in railways. Using GPS and GSM modules, an alert SMS containing the geographic coordinates of the damaged track is sent to the nearby railway authority, enabling them to take necessary steps to resolve the problem before any major accident occurs.

- [4] **“A Survey on Crack Detection Techniques in Railway Tracks, Maneesh Kumar M, Muthu S Murali, Saranya M, Arun S, Jayakrishnan R P, 2022”**

This paper provides an overview of various automatic railway track crack detection systems using robots. These systems typically include ultrasonic sensors, GPS, GSM modules, and Arduino-based crack detection, which are cost-effective and facilitate better safety standards

in railways. By using GPS and GSM modules, an alert SMS with the geographic coordinates of the damaged track is sent to the nearby railway authority to promptly address the issue before any major accidents occur.

[5] “Selah-Ajmer Train Derailment October 2023 Hindustan Times”

A tragic train derailment occurred in October 2023 in Vizianagaram district in Andhra Pradesh, India, involving the Sealah-Ajmer Express. Such derailments often result from undetected track problems like cracks or objects on the rails. This highlights the importance of proactive track inspection systems like the one described. By employing advanced sensors and real-time alerts, such a system could have potentially prevented this accident by identifying the issue beforehand.

2.2 Outcomes of literature survey

The literature surveys conducted shed light on the pressing issue of railway track safety and the importance of efficient crack detection systems. Railway transportation serves as a vital mode of public transit, offering fast and affordable travel options. However, the occurrence of accidents due to track defects like cracks poses a significant threat to passenger safety and infrastructure integrity. Various research efforts focus on developing advanced crack detection technologies to address this challenge. These include IoT-based systems, wireless security arrangements, and autonomous robots equipped with ultrasonic sensors and GPS/GSM modules. These innovations aim to detect cracks accurately and promptly, allowing for timely intervention to prevent potential accidents. The proposed solutions not only detect cracks but also provide real-time location information to railway authorities, enabling swift response measures. By leveraging advanced technologies, such as Arduino-based crack detection systems and autonomous robots, these approaches offer cost-effective and efficient alternatives to traditional inspection methods.

Overall, the outcomes of the literature surveys highlight the urgent need for effective crack detection techniques in railway infrastructure maintenance. By implementing innovative solutions, such as those proposed in the surveyed studies, railway operators can enhance safety standards, reduce the risk of accidents, and ensure the smooth operation of railway networks even in challenging environmental conditions.

CHAPTER 3

OBJECTIVE OF PROPOSED METHODOLOGY

The objective of the proposed methodology is to enhance the safety and efficiency of the Indian railway system by leveraging Internet of Things (IoT) technology to automatically detect track fractures and notify authorities promptly. The key objectives include:

- **Automated Detection of Track Fractures:** Implementing an IoT-based solution using ultrasonic sensors to continuously monitor railway tracks for cracks or fractures.
- **Real-time Location Tracking:** Utilizing GPS receivers to accurately track the location of detected track fractures in real-time.
- **Immediate Notification:** Integrating GSM modules to send instant messages or alerts to railway authorities as soon as a crack is detected, enabling prompt response and preventive measures.
- **Surveillance and Data Analysis:** Deploying cameras along the railway tracks to provide live video data, enabling authorities to visually analyze the extent of the rupture and take appropriate action.
- **Enhanced Safety Measures:** Providing timely information about track fractures helps in preventing accidents such as rail derailments, ensuring passenger and crew safety.
- **Efficient Maintenance Planning:** By collecting data on track fractures and their locations, railway authorities can plan maintenance activities more efficiently, reducing downtime and operational disruptions.
- **Cost-effective Solution:** Implementing IoT-based track fracture detection reduces the need for manual inspections, leading to cost savings in maintenance and operational expenses.

Overall, the proposed methodology aims to leverage IoT technology to improve the accuracy and timeliness of track fracture detection, thereby enhancing the safety, reliability, and operational efficiency of the Indian railway system.

CHAPTER 4

HARDWARE AND SOFTWARE

4.1 Hardware

4.1.1 Arduino Uno Microcontroller

Arduino comprises of both a physical programmable circuit board (commonly known as a microcontroller) and a programming software, or IDE (Integrated Development Environment) that can be run on a PC, used to compose and transfer PC code to the circuit board. It can be done by using the Arduino programming language (based on Wiring), and the Arduino Software (IDE), based on Processing. Unlike other programmable circuit boards, the Arduino does not require a different equipment (called a software engineer) to upload code to the circuit board, one can essentially utilize a USB link. Also, the Arduino IDE utilizes a rearranged rendition of C++, making it simpler to figure out how to program. In a word, Arduino make the functions of the micro-controller into a more accessible package. The Uno is one of the more prevalent boards in the Arduino family and an extraordinary option for the beginners.

Common Components of Arduino Boards:

There are different types of Arduino boards for different purposes. But all the boards have the majority of following components in common.

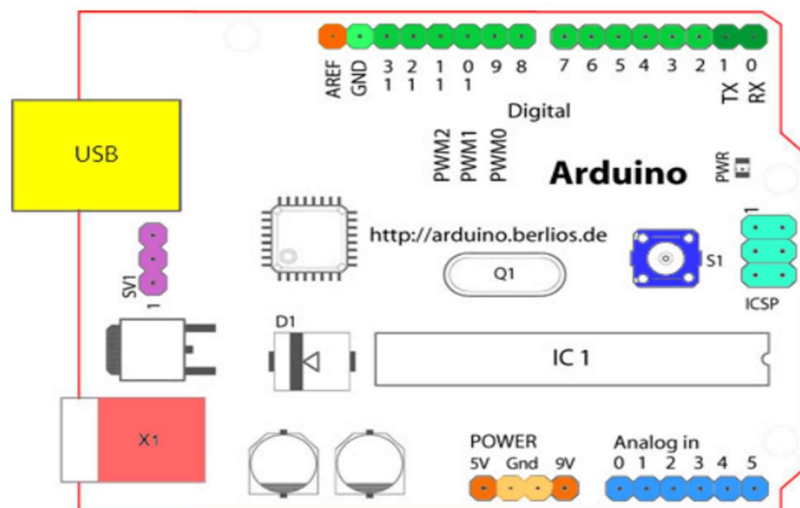


Fig 4.1: Common components of Arduino Board

Starting clockwise from the top center:

- Analog Reference pin (orange)
- Digital Ground (light green)
- Digital Pins 2-13 (green)
- Digital Pins 0-1/Serial In/Out - TX/RX (dark green) - These pins cannot be used for digital i/o (digitalRead and digitalWrite) if serial communication is also being used (e.g., Serial.begin).
- Reset Button - S1 (dark blue)
- In-circuit Serial Programmer (blue-green)
- Analog In Pins 0-5 (light blue)
- Power and Ground Pins (power: orange, grounds: light orange)
- External Power Supply In (9-12VDC) - X1 (pink)
- Toggles External Power and USB Power (place jumper on two pins closest to desired supply) - SV1 (purple)
- USB (used for uploading sketches to the board and for serial communication between the board and the computer; can be used to power the board) (yellow).

Digital Pins

The digital pins on an Arduino board can be used for general purpose input and output via the pinMode(), digitalRead(), and digitalWrite() commands. Each pin has an internal pull-up resistor which can be turned on and off using digitalWrite() (w/ a value of HIGH or LOW, respectively) when the pin is configured as an input.

- Serial: 0 (RX) and 1 (TX). Used to receive (RX) and transmit (TX) TTL serial data.
- External Interrupts: 2 and 3. These pins can be configured to trigger an interrupt on a low value, a rising or falling edge, or a change in value.
- PWM: 3, 5, 6, 9, 10, and 11. Provide 8-bit PWM output with the analogWrite() function. On boards with an ATmega8, PWM output is available only on pins 9, 10, and 11.

Analog Pins

The analog input pins support 10-bit analog-to-digital conversion (ADC) using the analogRead() function. Most of the analog inputs can also be used as digital pins: analog input 0 as digital pin 14 through analog input 5 as digital pin 19.

Power Pins

- 9V: The input voltage to the Arduino board when it's using an external power source (as opposed to 5 volts from the USB connection or other regulated power source). Different boards accept different input voltages ranges.
- 5V: The regulated power supply used to power the microcontroller and other components on the board. This can come either from VIN via an on-board regulator, or be supplied by USB or another regulated 5V supply.
- 3V3: (Diecimila-only) A 3.3volt supply generated by the on-board FTDI chip.
- GND: Ground pins

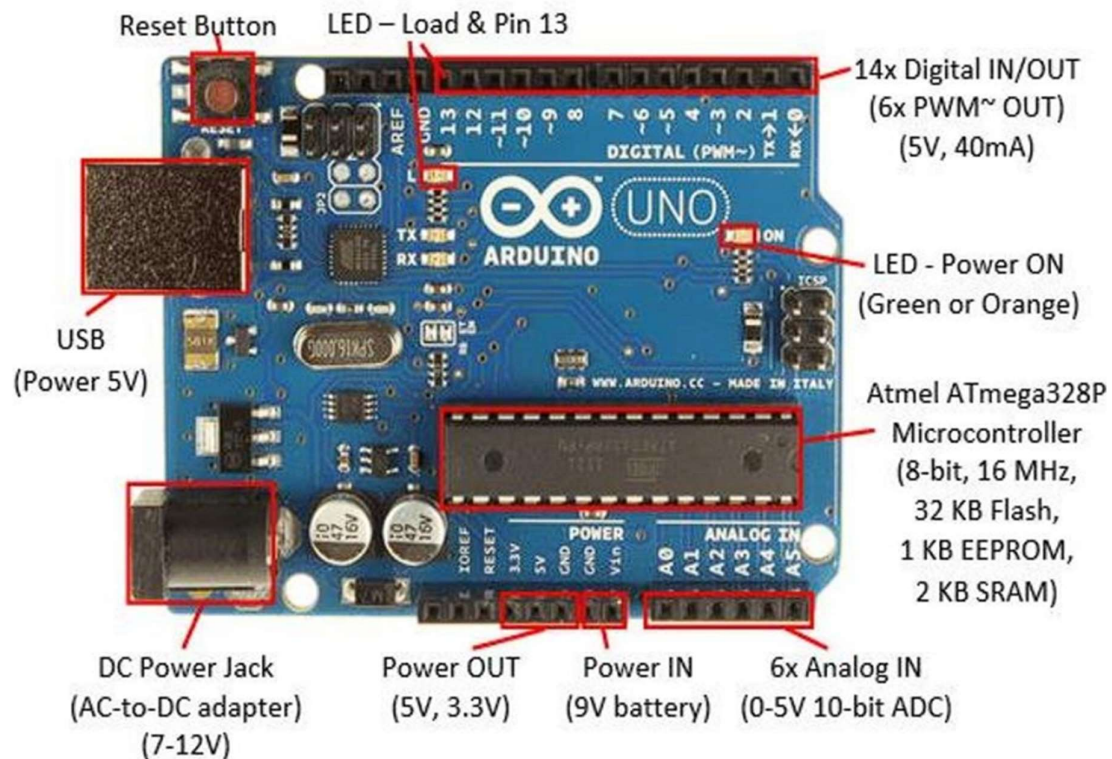


Fig 4.2: Arduino UNO

4.1.2 GSM Module

The acronym GSM is presently understood to mean Global System for Mobile Communications. We chose the SIM 800C because that support the 2G network. It requires a SIM (Subscriber Identity Module) card just like mobile phones to activate communication with the network. Also, they have

IMEI (International Mobile Equipment Identity) number similar to mobile phones for their identification. The MODEM needs AT commands, for interacting with processor or controller, which are communicated through serial communication. These commands are sent by the controller/processor. The modem sends back a result after it receives a command. Different AT commands supported by the modem can be sent by the processor/controller/computer to interact with the GSM and GPRS cellular network.

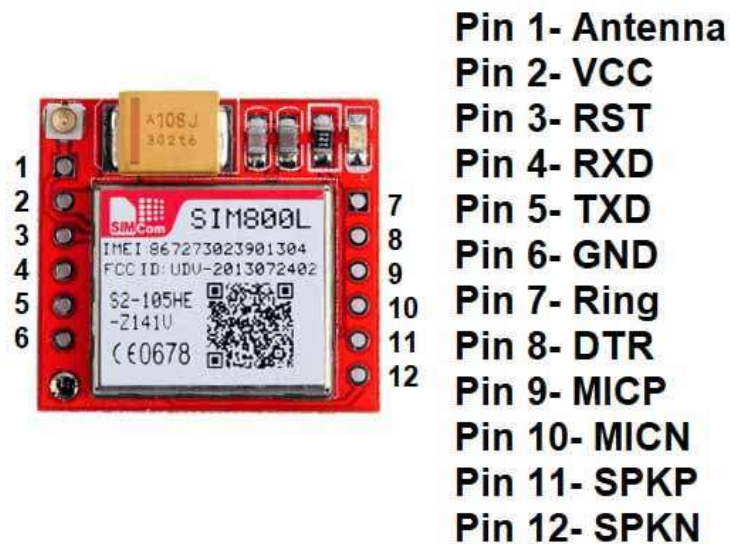


Fig 4.3: SIM800L GSM module

4.1.3 Buzzer

For alarm purposes a lot of electric bells, alarms and buzzers are available in the market that has got different prices and uses. The buzzer being used in this project is a 5-12 V buzzer and has got enough alarm sound to be used in a fire alarm system. Louder buzzer would have been even better but then their operating voltages are high as we had a supply of maximum up to 12V available with us on the board.



Fig 4.4 Buzzer

Table : Technical Specification of Buzzer

Operating Volatge(VDC)	5
Pin Spacing(mm)	6
Material	Plastic
Color	Black
Width(mm)	12(Body Diameter)
Height(mm)	8
Weight(gm)	1(approx.)
Shipment Dimensions	8x5x5 cm

4.1.4 DC Motor

DC motors consist of a stator, which generates a magnetic field, and a rotor, typically comprising coils or permanent magnets. When an electric current is supplied to the motor, interaction between the magnetic fields induces rotational movement in the rotor. This rotational motion finds application across a broad spectrum of industries, from robotics and automotive systems to household appliances and industrial machinery. With the ability to control speed and direction through varying voltage or pulse-width modulation, DC motors remain indispensable in modern engineering, powering everything from simple toys to complex automation systems.

**Fig 4.5: DC Motor**

Table : Technical Specification of DC Motor

Gear Material	Plastic
Operating Voltage (VDC)	12
Rated Torque(kg-cm)	0.5
Stall Torque(kg-cm)	2.0
Load Current Max(mA)	300
No-Load Current(mA)	60
Gearbox Diameter (mm)	37
Motor Length(mm)	75
Motor Diameter (mm)	32
Weight(gm)	80

4.1.5 Motor Driver

The LM298 is a versatile dual H-bridge motor driver IC commonly used for controlling DC and stepper motors. It can handle currents up to 2A per channel and includes built-in protection diodes to safeguard against back EMF. With two H-bridge outputs, it allows bidirectional control of two DC motors or one stepper motor. Its robust design, thermal shutdown protection, and ease of integration make it popular for robotics, automation, and motor control applications.

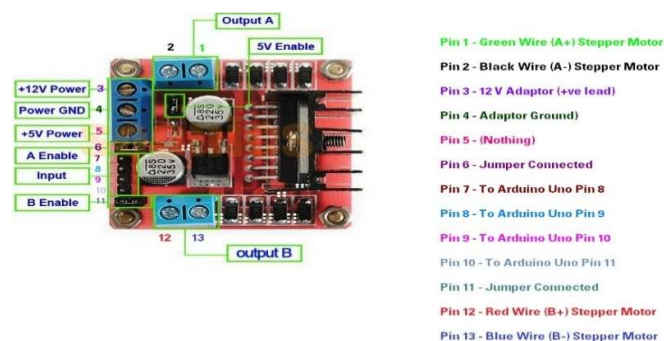
**Fig 4.6: Motor Driver**

Table : Technical Specification of Motor Driver

Double H Bridge Drive chip	L298N
Logical Voltage	5V
Drive Voltage	5V -35V
Logical Current	0-36 mA
Driver Current	2A (Max Single bridge)
Max Power	25 W
Dimension	43 * 43 * 26 mm
Weight	26g

4.1.6 PWM (Pulse width Modulation)

A PWM (Pulse Width Modulation) controller is a device that regulates the speed of a motor by varying the average voltage supplied to it. It achieves this by rapidly switching the power on and off at a fixed frequency, but with a variable duty cycle. The duty cycle represents the proportion of time the power is on (high) compared to the total cycle time. By adjusting the duty cycle, the effective voltage reaching the motor is changed, altering its speed. PWM controllers are commonly used in motor control applications due to their efficiency, simplicity, and ability to control speed smoothly and accurately. They find wide application in industries such as robotics, automotive, and manufacturing, where precise speed control is essential for optimal performance.

**Fig 4.7:** PWM Controller

4.1.7 Wheels

Wheels are essential components in vehicles and various mechanical systems, providing mobility, support, and traction. They typically consist of a circular frame with a central hub, designed to rotate freely around an axle. The outer surface of the wheel, known as the tire, often features treads or patterns to enhance grip and traction on different surfaces. Wheels come in a variety of sizes, materials, and designs to suit specific applications, ranging from small caster wheels on furniture to large, heavy-duty wheels on trucks and industrial machinery. The choice of wheel design depends on factors such as load capacity, terrain, speed, and environmental conditions. Overall, wheels play a crucial role in enabling motion and maneuverability across a wide range of vehicles and equipment, from bicycles and cars to airplanes and earthmoving machinery.



Fig 4.8: 45mm Wheels

4.2 Software Requirements

In our project we used a software for writing the code

- Arduino IDE – to write the code

4.2.1 Software Design

The program creation (development) cycle to create a C program:

- Do the requisite mental work. This is the most important part.
- Create the C source code. This can be done using a text editor, but is normally done within the IDE (Integrated Development Environment). C source files are plain text and saved with a “.c” extension.
- Compile the source code. This creates an assembly output file. Normally, compiling automatically fires up the assembler, which turns the assembly file into a machine language

output file.

- Link the output file with any required libraries using the linker. This creates an executable file. For desktop development, this is ready to test.
- For embedded development, download the resulting executable to the target hardware (in our case, the Arduino development board). For the Arduino, steps 3, 4, and 5 can be combined by selecting “Build” from the IDE menu.
- Test the executable. If it does not behave properly, go back to step one.

A program is text that you write using a programming language that contains behaviour that you need a processor to acquire. It basically creates a way of handling inputs and producing outputs according to these behaviours. Designing a program is the fact you have to think about first, before you begin coding it. It generally involves writing, drawing, and making schematics of all the actions you want your processor to make for you. Sometimes, it also implies to write what we call pseudo code. I hope you remember that this is what we created in the previous chapter when we wanted to define precisely all the steps of our desired LED behaviour. Writing a program is typically what converts the pseudo code into real and well-formed code. It involves having knowledge of programming languages because it is the step when you really write the program. This is what we'll learn in a moment. Testing is the obvious step when you run the program after you made some modifications to the code. It is an exciting moment when you also are a bit afraid of bugs, those annoying things that make running your program absolutely different from what you expected at first. Debugging is a very important step when you are trying to find out why that program doesn't work well as expected. You are tracking typo errors, logic discrepancies, and global program architecture problems. You'll need to monitor things and often modify your program a bit in order to precisely trace how it works. Maintaining the source code is the part of the program's life that helps to avoid obsolescence. The program is working and you improve it progressively; you make it up-to-date considering hardware evolutions, and sometimes, you debug it because the user has this still undiscovered bug. This step increases the life duration of your program. Programming is the process of designing, writing, testing, debugging and maintaining the source code of computer programs.

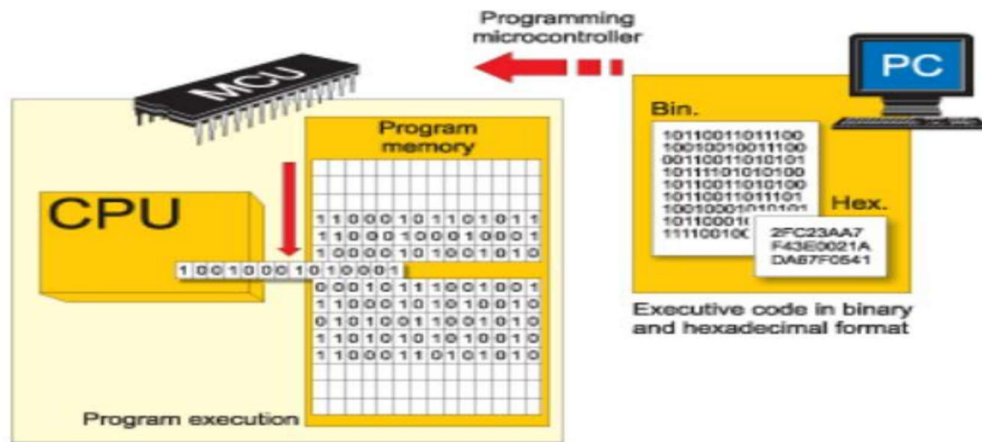


Fig 4.9: Program creation steps

The microcontroller executes the program loaded in its Flash memory. This is the so-called executable code comprised of seemingly meaningless sequence of zeros and ones. It is organized in 12-, 14- or 16-bit wide words, depending on the microcontroller's architecture. Every word is considered by the CPU as a command being executed during the operation of the microcontroller. For practical reasons, as it is much easier for us to deal with hexadecimal number system, the executable code is often represented as a sequence of hexadecimal numbers called a Hex code. It used to be written by the programmer. All instructions that the microcontroller can recognize are together called the Instruction set.

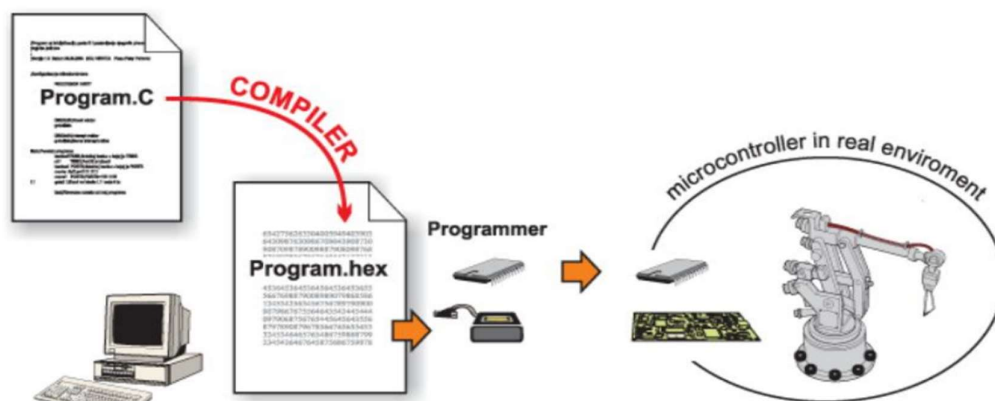


Fig 4.10: Generating hex file

Every sketch needs two void type functions that do not return any value, `setup()` and `loop()`. The `setup()` method is run once, just after the Arduino board is powered up and the `loop()` method is run continuously afterwards. The `setup()` is where you want to do any initialization steps, and `loop()` is for codes you want to run over and over again to program ATMEGA328P. At first, we have to burn

boot loader of new ATMEGA328P using ARDUINO-UNO R3 Programmer. For burn boot loader connect new ATMEGA328P with programmer. Connection is shown below.

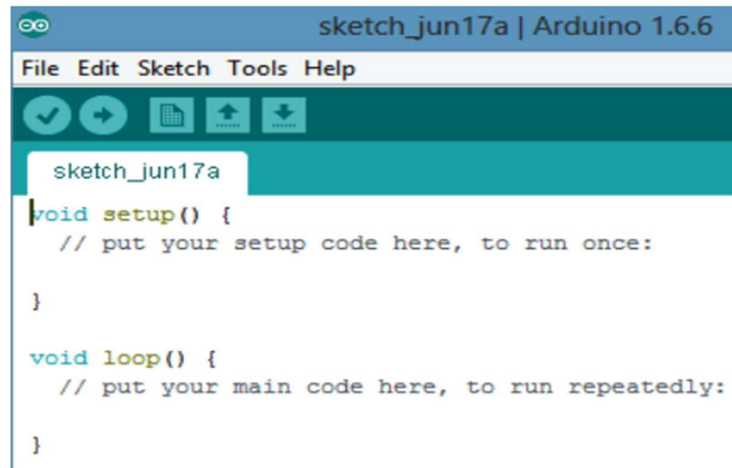


Fig 4.11: Arduino sketch simple function

Caution: Make sure Programming kit is not connected to external power source because it gets required power from USB cable connected to computer and circuit with ATMEGA328P have regulated +5V supply.

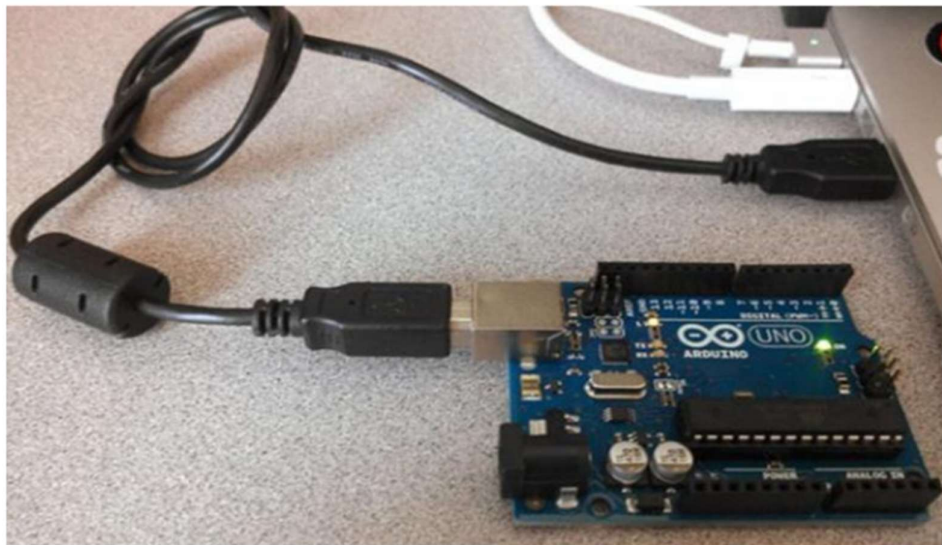


Fig 4.12: Interfacing ARDUINO Programmer with PC

Then this System is ready for burn boot loader, to do this action simply open IDE on computer then upload configuration code which is consists of two different codes named “optiLoader.h” and “optiLoader.pde”. Then upload both code to microcontroller, process is shown below.

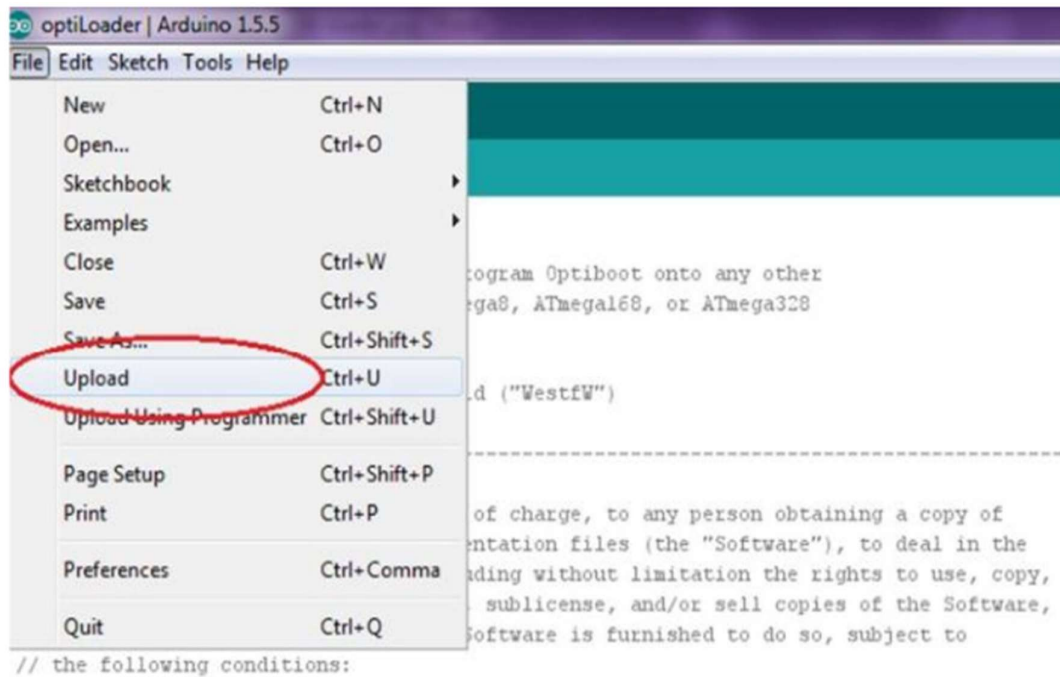


Fig 4.13: Upload Boot loader to ARDUINO Programmer

4.3 System Testing

First of all, all the hardware units of the system were tested and it was ensured that in a good working condition or not. Then each and every unit were interfaced and implemented individually with the microcontroller board and drove with the software according to the necessity of the application. The testing of the application was not done at once after it was completed. Rather each unit of the application was tested individually. The second unit was not tested until the first unit gave the expected result and until it was not working according to the necessity of the application. After all of the units were working correctly, the units were kept together and then the whole system was developed and tested.

It was easy to figure out the bugs and the problem of the system as the behavior of each unit was known while testing it. It would be impossible to figure out the system development when a sensor gets activation and defined program and the response at the output pin.

We also used in this circuit to light ON an indication LED with connected a 220Ω resistor output pin to common GND. Output pin D5 is common for the system as we defined program to activate a buzzer alarm and send alert message.

CHAPTER 5

IMPLEMENTATION AND METHODOLOGY

5.1 Block Diagram

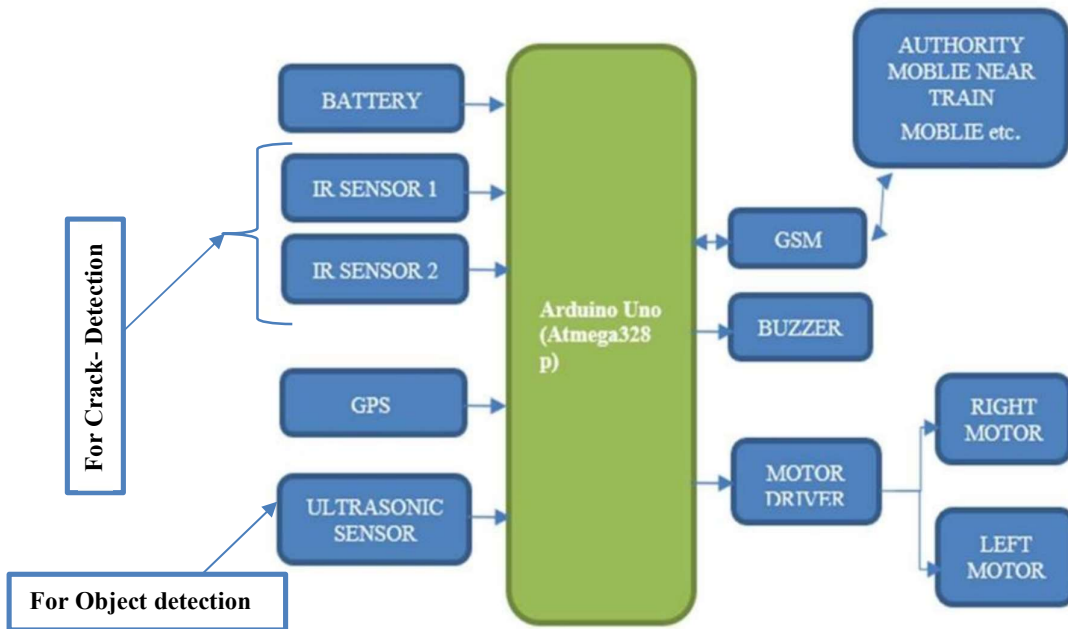


Fig 5.1: Block Diagram of Overall System

Fig 5.1 shows the block diagram of “Railway track Crack detection and object detection using GSM & GPS”. The system consists of Arduino Uno, LCD Display, GPS, GSM, Ultrasonic sensor, IR sensors, buzzer, battery source and DC Motor. The Arduino Uno microcontroller, which acts as a brain of the system. This microcontroller controls the circuit function. Various components are interfaced with this microcontroller. In this system we have interfaced two IR sensors with the microcontroller to check damage or crack present in the track of the railway line. ultrasonic sensor is used to detected the objects present on the tracks. A GPS receiver is also interfaced with the microcontroller to determine the exact location of the crack on the railway track. Then through by using GSM microcontroller will send the message for near railway station and main authority. Two DC motors are used to move the robot in forward direction. The architecture of the proposed system also consists of a camera, interfaced with the microcontroller for monitoring railway track using vedio analysis , and a buzzer to generate beep sound after detecting any crack or object.

5.2 Implementation of Proposed Methodology

Fig 5.2 shows block diagram of “Railway track crack and object detection using IOT”, there are two set of IR sensor units fixed to the front side of the vehicle with the microcontroller to check the crack present in the track of the railway line. When the vehicle is switched on, it moves forward along the track. The IR sensors check the condition of the tracks. In normal condition the motor, LDR, Serial transmission is in initial stage. When the power supplies the microcontroller then it starting the motor in forward direction and sends the messages to the microcontroller using serial transmission.

When the crack is detected by the IR sensor automatically vehicle stops, and the GPS receiver triangulates the position of the vehicle to receive the Latitude and Longitude coordinates of the vehicle position, from satellites and Latitude and Longitude coordinates received by GPS are converted into a text message which is done by microcontroller. The GSM Module sends the text message on Mobile Phone.

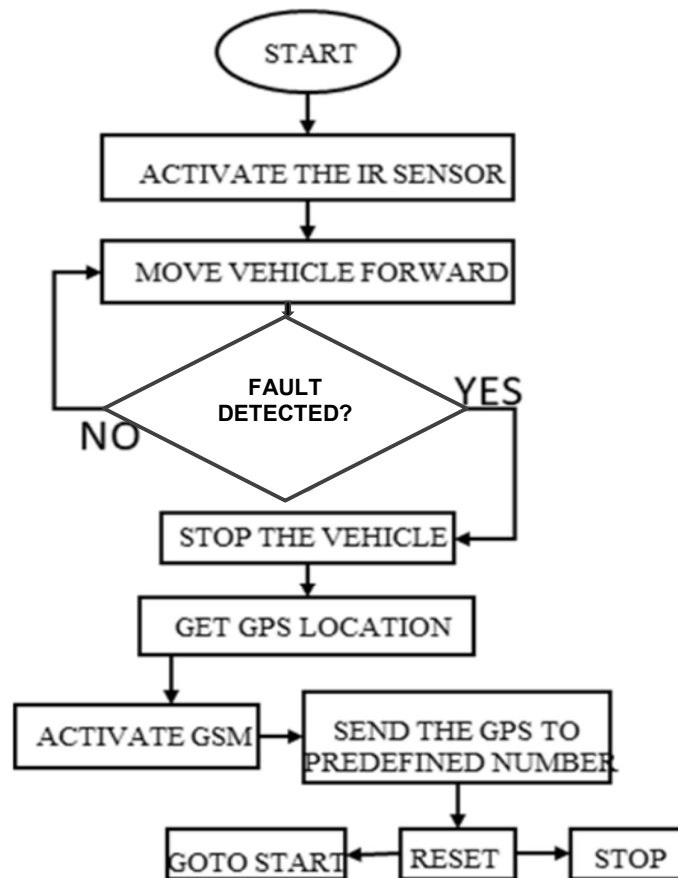


Fig 5.2: Flow Chart

At Normal Condition:

The IR transmitter sensor is transmitting the infrared rays and received by the IR receiver. At normal condition Transistor is OFF condition, at the same time relay is also OFF, so that the vehicle moving forward continuously.

At Crack Condition:

At crack detection conditions the IR transmitter transmit infrared rays and received by the IR receiver after hitting the object. If there is no crack present on the track then output of IR LED and Photodiode will be high.

If the crack is present on the track, is detected by the system, the sensor reflection will be equal to zero and the robot will be stopped automatically. When this output is high then it is considered that there is no crack on the track. But if any crack is detected by the sensor the output of the sensor given to the microcontroller will be zero and again the microcontroller will stop the robot. That means when a crack is detected by the IR sensor the vehicle stops at once, and the GPS receiver sends the position of the vehicle to receive the Latitude and Longitude coordinates of the vehicle position, from satellite as show in the Fig 5.2.

5.3 Implementation and Design

5.3.1 Hardware Implementation

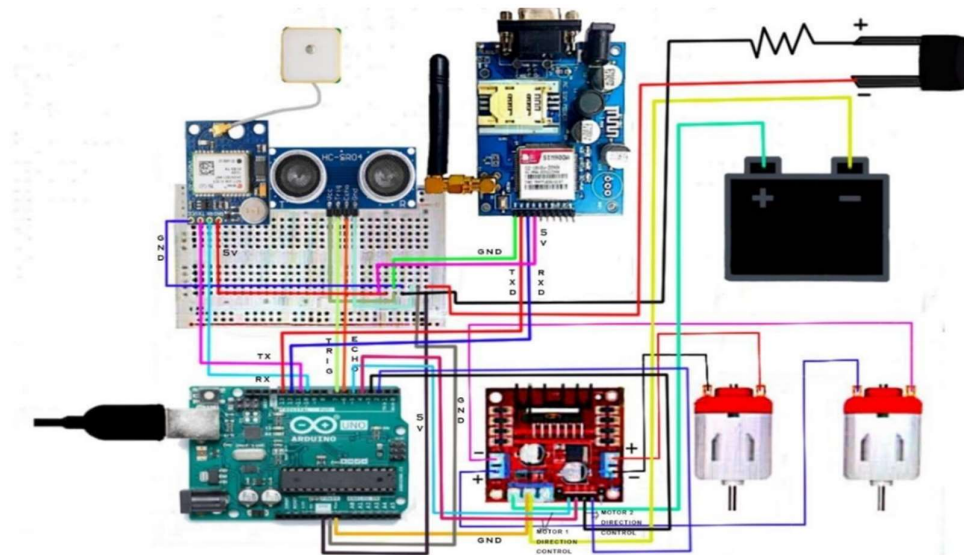


Fig 5.3: Object detection circuit diagram

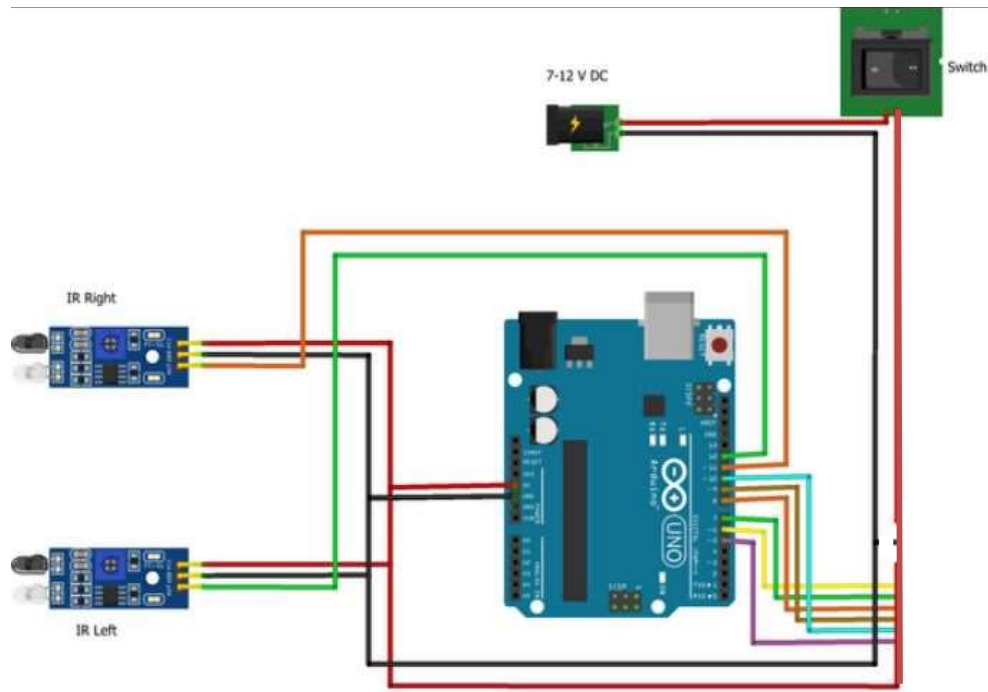


Fig 5.4: Crack Detection circuit diagram

- Railway Crack Detection Circuit using IR Sensor:**

One approach to railway crack detection utilizes an infrared (IR) sensor. This method involves positioning two IR emitter-receiver pairs facing each other across the rail. When the track is smooth, the IR light emitted by one sensor travels directly back to the receiver on the opposite side. However, if a crack is present, the uneven surface disrupts the light path. This disruption causes less light to reach the receiver, resulting in a weaker signal.

The microcontroller unit (MCU) continuously monitors the output signal from the IR receiver. A significant decrease in the received signal compared to the normal level might indicate a crack. This method offers a simple solution, but it has limitations. External factors like dirt, debris, or even uneven track surfaces can also affect the IR signal, potentially leading to false alarms. Additionally, the effectiveness of this method might be limited by the depth and width of cracks that can be detected.

- Railway Object Detection Circuit using Ultrasonic Sensor:**

A more common and reliable approach for railway object detection utilizes ultrasonic sensors. The circuit employs an ultrasonic sensor that emits a high-frequency sound wave. This sound wave travels and reflects off any object in its path, including objects

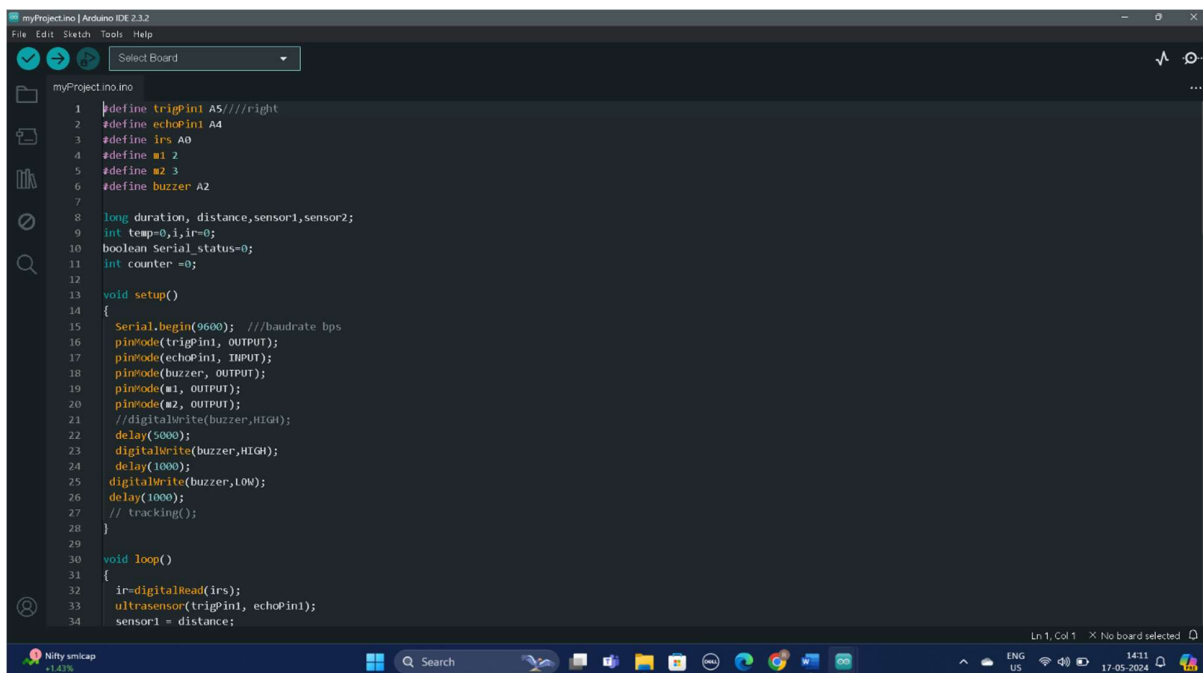
on the railway track. The reflected sound wave is then received by the sensor.

The microcontroller unit (MCU) plays a crucial role in this system. It measures the time it takes for the sound wave to travel and return. Based on the speed of sound and the round-trip time, the MCU can calculate the distance between the sensor and the object. If the calculated distance falls below a pre-set threshold (indicating an object is close to the sensor), the MCU triggers an alert.

Compared to using an IR sensor for crack detection, this ultrasonic method offers a more reliable way to detect objects on the tracks. However, ultrasonic sensors also have limitations. Potential interference from other sound sources or objects with complex shapes that might scatter the sound waves can affect their accuracy.

5.3.2 Software Implementation

As shown in the Fig 5.5, the code is written on the Arduino IDE and python code is for web-interface result on Vs-code IDE software which is shown in Fig 5.6. Arduino code is to be uploaded to the microcontroller. The code is dumped into Arduino board by connecting it to the port of the system as shown in fig. 4.13.



```

1  #define trigPin1 A5///right
2  #define echoPin1 A4
3  #define irs A0
4  #define m1 2
5  #define m2 3
6  #define buzzer A2
7
8  long duration, distance,sensor1,sensor2;
9  int temp=0,i,ir=0;
10 boolean Serial_status=0;
11 int counter =0;
12
13 void setup()
14 {
15   Serial.begin(9600); //baudrate bps
16   pinMode(trigPin1, OUTPUT);
17   pinMode(echoPin1, INPUT);
18   pinMode(buzzer, OUTPUT);
19   pinMode(m1, OUTPUT);
20   pinMode(m2, OUTPUT);
21   //digitalWrite(buzzer,HIGH);
22   delay(5000);
23   digitalWrite(buzzer,HIGH);
24   delay(1000);
25   digitalWrite(buzzer,LOW);
26   delay(1000);
27   // tracking();
28 }
29
30 void loop()
31 {
32   ir=digitalRead(irs);
33   ultrasensor(trigPin1, echoPin1);
34   sensor1 = distance;

```

Fig 5.5: Arduino Code as written on the Arduino IDE

```

1 from flask import Flask, render_template, request
2 import cv2
3 import numpy as np
4 import base64
5 import math
6 from datetime import datetime
7
8 app = Flask(__name__)
9
10 # Function for oriented non-max suppression
11 def orientated_non_max_suppression(mag, ang):
12     ang_quant = np.round(ang / (np.pi / 4)) % 4
13     winE = np.array([[0, 0, 0], [1, 1, 1], [0, 0, 0]])
14     winSE = np.array([[1, 0, 0], [0, 1, 0], [0, 0, 1]])
15     winS = np.array([[0, 1, 0], [0, 1, 0], [0, 1, 0]])
16     winSW = np.array([[0, 0, 1], [0, 1, 0], [1, 0, 0]])
17
18     magE = non_max_suppression(mag, winE)
19     magSE = non_max_suppression(mag, winSE)
20     magS = non_max_suppression(mag, winS)
21     magSW = non_max_suppression(mag, winSW)
22
23     mag[ang_quant == 0] = magE[ang_quant == 0]
24     mag[ang_quant == 1] = magSE[ang_quant == 1]
25     mag[ang_quant == 2] = magS[ang_quant == 2]
26     mag[ang_quant == 3] = magSW[ang_quant == 3]
27
28     return mag
29
30 # Function for non-max suppression
31 def non_max_suppression(data, win):
32     # convert kernel to cv_8U
33     win = win.astype(np.uint8)
34
35     data_max = cv2.dilate(data, win)

```

Fig 5.6: Python Code as written on the Vs-Code for Detailed report

Code after compilation is dumped into the microcontroller for the track detection and object detection. After successful compilation of the code and dumping, power is supplied to the model to start the functioning of the proposed system.

5.4 Applications

Railway track crack detection and object detection with GPS and GSM using IoT offers a significant advancement in railway safety. Here are some key applications of this system:

1. Proactive Track Maintenance:

- By continuously monitoring the tracks for cracks and objects, the system allows for early detection of potential problems. This proactive approach prevents issues from escalating into major failures that could cause derailments.
- Early detection enables timely repairs and maintenance, minimizing track downtime and disruptions to railway schedules.

2. Improved Response to Incidents:

- Upon detecting a crack or object, the system transmits a real-time alert with GPS location data. This allows railway authorities to respond swiftly and efficiently.
- Knowing the precise location of the problem eliminates the need for manual searches,

leading to faster repairs and reduced risk of accidents.

3. Enhanced Safety and Reduced Accidents:

- The primary goal of this system is to prevent train accidents caused by undetected railway track cracks and obstacles. By proactively addressing these issues, the system significantly enhances overall railway safety.
- Real-time alerts and swift response times minimize the risk of accidents, safeguarding passengers, personnel, and cargo.

4. Data Collection and Analytics:

- The system can be designed to collect and store sensor data along with GPS coordinates over time. This data can be analyzed to identify trends in crack development or areas prone to object placement.
- This data analysis allows for the development of preventive maintenance strategies, focusing resources on areas with higher risks.

5. Remote Monitoring and Improved Efficiency:

- The system enables remote monitoring of railway tracks, eliminating the need for constant manual inspections. This translates to reduced labor costs and improved operational efficiency.
- Real-time data access allows for better decision-making regarding track maintenance.

Additional Applications:

- The system can be integrated with existing railway infrastructure to provide real-time updates on track conditions to train operators.
- Integration with automated warning systems can trigger alerts for trains approaching a detected crack or object.

5.5 Advantages

- **Real-Time Alerts and Response:** Upon detecting an issue, the system transmits a real-time alert with precise GPS location via GSM. This allows railway authorities to respond swiftly and efficiently to the exact location of the problem.
- **Reduced Downtime and Improved Efficiency:** Early detection of problems minimizes the need for extensive repairs and lengthy track closures. This translates to smoother railway operations and improved efficiency.

- **Data Collection and Analytics:** The system can collect and store sensor data along with GPS coordinates. Analyzing this data can help identify trends in crack development or areas prone to object placement, enabling targeted preventive maintenance strategies.
- **Remote Monitoring and Cost Savings:** The system allows for remote track monitoring, potentially reducing the need for constant manual inspections, leading to lower labour costs.

5.6 Disadvantages

- **Sensor Performance Degradation:** Harsh weather conditions like extreme temperatures or electromagnetic interference might affect sensor performance, potentially leading to missed detections or false alarms.
- **Cellular Network Connectivity Problem:** Reliable cellular network connectivity is crucial for uninterrupted operation. In remote areas with weak or no cellular signal, the system might be compromised.
- **Maintenance and Expertise Overhead:** The system requires ongoing maintenance to ensure proper function. Technical expertise might be needed to troubleshoot any issues that arise.

CHAPTER 6

RESULTS AND DISCUSSION

6.1 Hardware Result

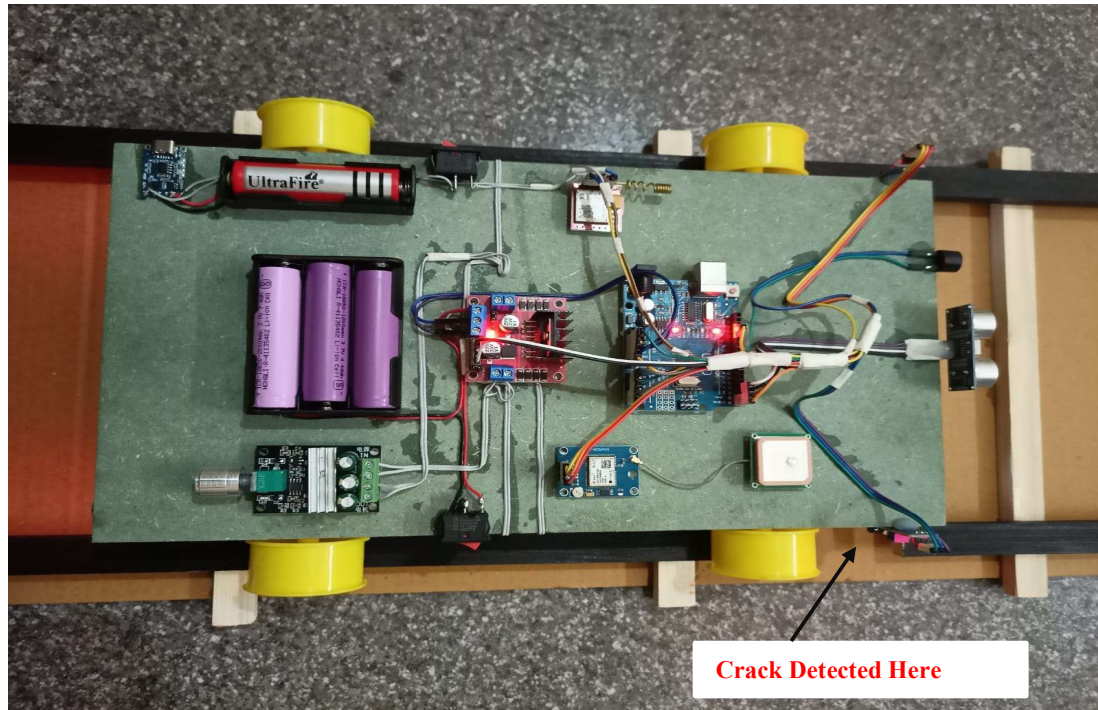


Fig 6.1: Hardware model of Railway track crack and object using IOT

The system consists of Arduino Uno, Ultrasonic sensor, Motor driver, PWM, GSM, GPS, IR sensors, UltraSonic sensor and DC Motor. The Arduino Uno microcontroller, which acts as a brain of the system. This microcontroller controls the circuit function. Various components are interfaced with this microcontroller. The hardware components used in this system requires regulated power supply for the operation. This power is provided by the rechargeable battery connected in the system. In this system we have interfaced two IR sensors and a Ultrasonic sensor with the microcontroller to check damage or crack and object present in the track of the railway line. A GPS receiver is also interfaced with the microcontroller to determine the exact location of the crack on the railway track. The SMS is used to send the message for railway station or main authority using GSM. Four DC motors with wheels are used to move the robot in forward direction. A wireless camera provides the live video to the device in which the application of that camera is installed. The architecture of the proposed system also consists of a buzzer, interfaced with the microcontroller for the alert purpose.

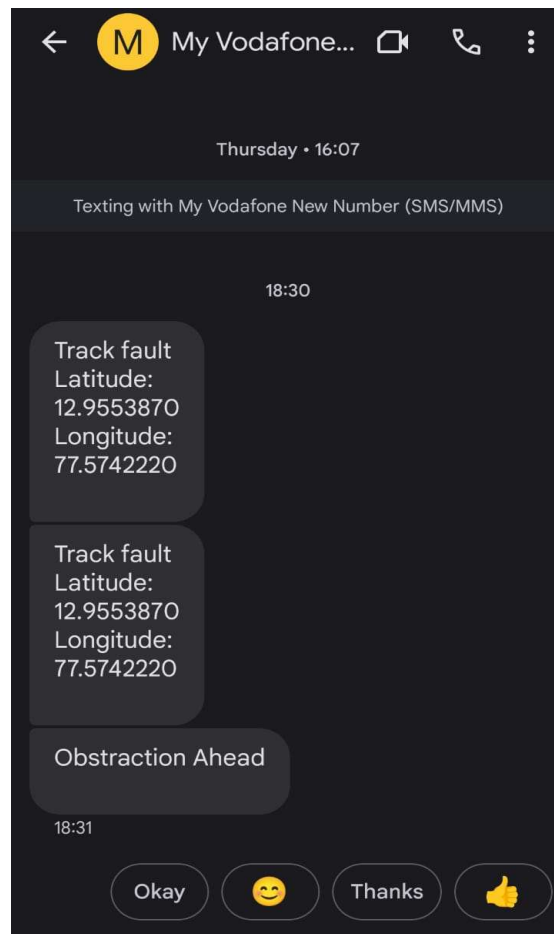


Fig 6.2: Sending SMS using GSM module

An IoT system utilizing GSM modules safeguards railway tracks by detecting cracks and objects. Sensors like ultrasonic or cameras scan the tracks, feeding data to a central microcontroller. This brain analyzes the data, identifying concerning cracks or objects on the tracks. If an issue arises, the microcontroller triggers the GSM module. Equipped with a SIM card, the GSM module connects to the cellular network and transmits an alert message.

This message, often an SMS, includes details like the problem type (crack or object), severity, and location data (likely from a built-in GPS module). The GSM module sends this alert to a predetermined phone number or server, allowing railway authorities to receive real-time notifications and take swift action, ensuring the safety of the railway system.

6.2 Software Result:

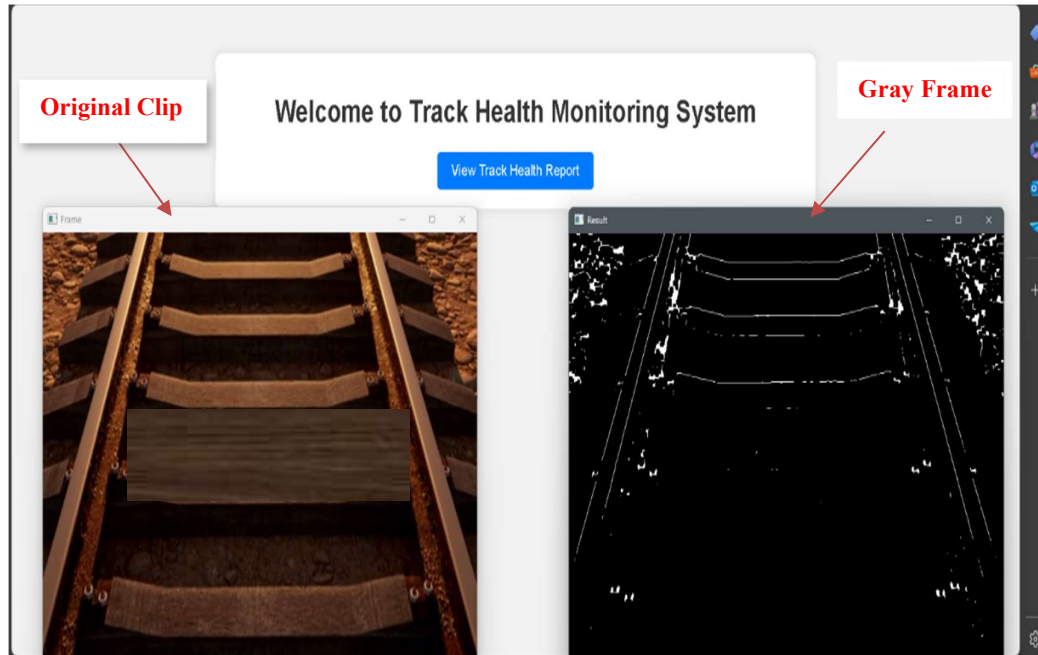


Fig 6.3: Analyzing Real video clip

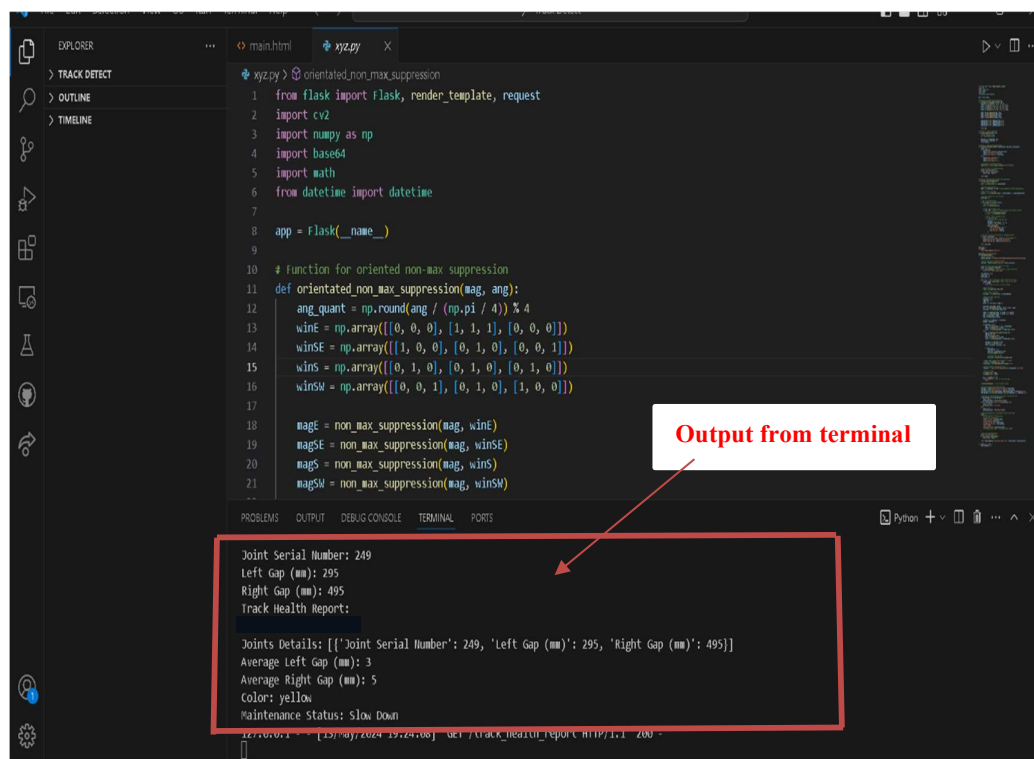


Fig 6.4: Result from Vs-code Terminal

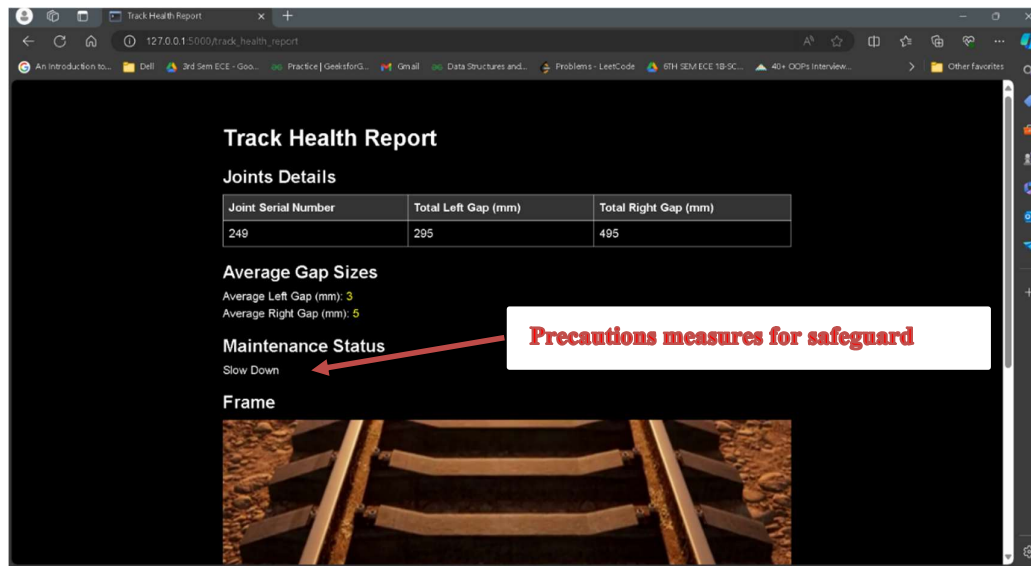


Fig 6.5: Detailed Report in web-interface

Our railway safety system transcends basic sensors by harnessing the power of cutting-edge computer vision. Cameras mounted on inspection vehicles or drones capture video footage of the tracks. This video data undergoes meticulous analysis using advanced techniques like edge detection and non-maximal suppression. These sophisticated algorithms meticulously examine each video frame, highlighting edges and eliminating redundant information. This allows the system to identify even the most subtle cracks on the track surface with unparalleled accuracy, surpassing the limitations of traditional sensor-based systems. But this model goes beyond just crack detection. It employs contour analysis to identify track joints the crucial connection points between rail segments. Analysing these joints provides valuable insights into the overall health of the railway infrastructure. This comprehensive assessment ensures no aspect of track integrity goes unnoticed. The system does not wait for problems to escalate.

When a crack is detected, an immediate alert is sent to concerned authorities via a user-friendly web interface. This real-time notification system eliminates delays and allows for a swift response. The web interface goes beyond just alerts; it empowers stakeholders with actionable data. Detailed reports showcase not only the size of the detected crack but also its precise location. For track joints, the reports include vital information like serial numbers and specific gap measurements on both sides. With this knowledge, they can prioritize maintenance needs and ensure timely interventions, preventing accidents and ensuring safe and efficient railway operations. Our system, with its advanced computer vision and real-time data delivery, represents a significant leap forward in railway safety and reliability.

6.3 Discussion

Current railway inspections often miss critical issues due to their manual nature. This discussion highlights a revolutionary IoT system for proactive track safety: crack and object detection with GPS and GSM. Unlike traditional methods, the system continuously scans tracks with sensors like ultrasonics or cameras, enabling early crack detection before they worsen. When a problem arises, the system acts intelligently. A central processing unit analyzes sensor data and triggers an immediate alert via a GSM module. This critical alert includes details like the problem type (crack or object), severity, and most importantly, the exact location from a GPS module. Railway authorities receive real-time notification, allowing for a swift response to the precise location. The benefits extend beyond immediate response. Early detection minimizes repairs and downtime, leading to smoother operations. This system paves the way for a safer and more efficient railway future, safeguarding lives and ensuring a reliable transportation network.

CHAPTER 7

CONCLUSION AND FUTURE WORKS

7.1 Conclusion

Railway safety hinges on the integrity of its tracks. Undetected cracks and obstacles can have devastating consequences. This innovative project tackles this challenge head-on with a robotic railway inspection system designed for proactive detection and swift response. The system employs a robotic vehicle equipped with high-tech sensors. These sensors, like ultrasonic detectors or vision-based cameras, continuously scan the tracks. This vigilant approach allows for the early identification of potential problems, well before they can develop into major issues. The collected sensor data is fed into a central processing unit, the brain of the system. This intelligent unit analyses the data in real-time, pinpointing concerning cracks exceeding a certain depth or objects present on the tracks. The system doesn't wait for problems to worsen. When an issue is detected, the GSM module, equipped with a SIM card, springs into action. It connects to the cellular network and transmits a critical alert message containing valuable details. This message typically includes the type of problem (crack or object), its severity (depth of crack, size of object), and most importantly, the exact location data retrieved from a built-in GPS module.

This real-time notification system ensures a prompt response from the maintenance crew. The alert, often an SMS message, is sent to pre-programmed phone numbers of designated railway authorities. Armed with the precise location provided by the GPS data, the crew can be dispatched immediately to the exact spot, allowing for swift repairs or removal of obstacles, significantly reducing the risk of accidents. The benefits extend beyond immediate response. By detecting and addressing problems early on, the system minimizes track closures and repairs. This translates to less downtime for the railway system, leading to smoother operations and improved efficiency. The ultimate goal is to prevent railway accidents caused by undetected cracks and obstacles. By implementing this robotic vehicle system, railways can proactively safeguard their infrastructure, ultimately saving lives and protecting passengers and personnel. This technology marks a significant leap towards achieving the highest safety standards for railway transportation.

7.2 Future works

- 1) **Machine Learning Algorithms:** Develop machine learning algorithms, possibly based on deep learning techniques, to analyze data from the sensors in real-time. These algorithms can detect and classify various types of cracks, objects, and potential hazards on the railway tracks.
- 2) **Data Analysis and Predictive Maintenance:** Analyze the collected data over time to identify trends and patterns in track degradation and object intrusion. Develop predictive maintenance models that anticipate potential failures or hazards based on historical data, enabling proactive maintenance measures.
- 3) **Integration with Maintenance Systems:** Integrate the crack and object detection system with existing railway maintenance systems to streamline maintenance workflows. This integration can facilitate task assignment, resource allocation, and maintenance scheduling based on real-time track condition data.
- 4) **Advancement in communication:** Advancement in communication for quality communication in rural areas is necessary for sending alert message via GSM.

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